
Energy-Guided Smoothness for Robust Graph Classification under Label Noise

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Abstract

Graph Neural Networks (GNNs) are vulnerable to label noise in graph classification, yet, unlike for image or node classification, the mechanisms that drive this vulnerability are poorly understood. We show that GNN robustness in this setting is governed by a property of the *learned representations*, not of the labels themselves: the within-graph Dirichlet energy of node features. Energy decreases while the model captures genuine class structure, but increases sharply, particularly in high-frequency components, when the model begins to memorize corrupted graph-level labels. We give a theoretical justification for why this decrease is non-trivial in graph classification by combining (i) a spectral argument that relates dataset-level energy to per-graph low-frequency structure and (ii) an entropy-based argument adapted from Qian et al. [2025], showing that uncontrolled energy growth disproportionately harms graphs that are most informative for classification. Building on this insight, we propose three complementary interventions, a spectral weight constraint (CE+W2), a Dirichlet-energy regularizer ($\mathcal{L}_{DE}$), and a robust loss function (GCOD), which all suppress harmful high-frequency energy through different mechanisms. Across eight benchmarks, including the full and modified ogbg-ppa, under symmetric noise (up to 60%), asymmetric noise, and a transfer to node classification on Cora, the three methods yield consistent gains while preserving clean-data performance. GCOD matches or exceeds the strongest noise-robust baselines (SOP, OMG) at $\sim 70\times$ lower training overhead, establishing Dirichlet energy as both a diagnostic signal and a principled design target for noise-robust GNNs.

1 Introduction

Graph Neural Networks (GNNs) are now the workhorse of *graph classification*, the task of assigning a single label to an entire graph, with applications spanning molecular property prediction [Wale and Karypis, 2006], bioinformatics [Borgwardt et al., 2005], and social-network analysis [Yanardag and Vishwanathan, 2015a]. In real deployments, the labels attached to graphs are routinely noisy: assays disagree, annotators err, and weak supervision injects systematic mistakes. Although learning under label noise has been thoroughly studied for images [Algan and Ulusoy, 2021] and for node classification [Dai et al., 2021, Yuan et al., 2023a], graph classification under noisy labels is dramatically less explored [NT et al., 2019, Yin et al., 2023].

Graph classification is a fundamentally different setting. In node classification, label noise corrupts *individual node labels* on a single graph, and homophily, the tendency of connected nodes to share labels, is a structural prior that directly couples Dirichlet energy of the *label signal* to noise. In graph classification, label noise flips the label of an *entire graph*: nodes have no individual

classification labels, and within-graph homophily/heterophily is largely orthogonal to the graph-level label. The standard tools and intuitions from node classification therefore do not transfer.

Our central observation is non-trivial. We measure the Dirichlet energy of *learned node representations*, not of labels. Empirically, this energy first decreases as the model captures genuine, transferable patterns. Then, when the model begins to memorize incorrect graph-level labels, the within-graph energy rises sharply, with the surge concentrated in the high-frequency spectrum. This is not a tautology: noise is injected only at the graph-label level, while node features and edges are unchanged. We show in Theorem 1 that, under a bi-Lipschitz assumption on the GNN, fitting noisy graph-level labels for two structurally similar graphs *forces* their within-graph Dirichlet energies to grow, via the Graph Poincaré inequality. The connection between graph-level label noise and within-graph spectral sharpening therefore emerges through training dynamics, not by definition.

Why does suppressing this energy growth help? Combining a dataset-level spectral identity (Proposition 1) with an entropy-based argument adapted from Qian et al. [2025], we show in Section 6 that uncontrolled within-graph energy growth disproportionately damages *high-entropy graphs*, the very graphs that carry the most discriminative signal at the pooling stage. A non-trivial decrease of within-graph Dirichlet energy thus simultaneously (i) prevents the network from fitting graph-level label noise and (ii) preserves the high-entropy regions on which graph classification accuracy depends.

Contributions.

- We give the first systematic study of *when* GNNs fit graph-level label noise (Section 4), isolating three failure modes: over-parameterization relative to task complexity, low-order graphs, and small training sets.
- We establish Dirichlet energy as a diagnostic for graph-level noise memorization (Section 5), with a frequency decomposition showing the surge is concentrated in high-frequency components.
- We provide *theoretical grounding for why a non-trivial energy decrease prevents noise fitting in graph classification* (Section 6), through a dataset-level spectral identity (Proposition 1), a Poincaré–bi-Lipschitz lower bound (Theorem 1), and an entropy-based asymmetry result (Corollary 1) showing that high-entropy graphs absorb the bulk of the energy growth. This is what makes the connection *non-trivial* for graph classification, in contrast to the trivial relationship that exists for node classification.
- We translate the theory into three convergent interventions (Section 7): spectral weight constraints (CE+W2), explicit Dirichlet regularization ($\mathcal{L}_{DE}$), and a centroid-based loss GCOD. The fact that mechanistically distinct interventions all reduce energy and improve robustness provides causal evidence beyond a single-method study.
- We evaluate on eight benchmarks under symmetric, asymmetric and high-rate (60%) noise, the full and modified ogbg-ppa, and a node-classification transfer (Cora). GCOD matches or exceeds OMG [Yin et al., 2023] and SOP [Liu et al., 2022] at $\sim 70\times$ lower compute, and our energy-based methods generalize beyond their original setting.

2 Related Work

Learning under label noise. Robust loss functions [Ghosh et al., 2017, Patrini et al., 2017], sample re-weighting [Liu and Tao, 2015, Liu et al., 2022], early-learning regularizers [Arpit et al., 2017], and small-loss filtering [Gui et al., 2021] dominate the image-classification literature. Wani et al. [2023] introduce NCOD, a centroid-based outlier-discounting loss that we adapt to graphs.

Noisy-label learning on graphs. Most prior work concerns *node* classification, exploiting homophily, contrastive losses, or pseudo-labels [Dai et al., 2021, Yuan et al., 2023a,b, Li et al., 2024, Kang et al., 2018]. For *graph* classification, NT et al. [2019] propose a surrogate loss that discards estimated-noisy labels under specific assumptions, and OMG [Yin et al., 2023] combines contrastive learning, MixUp [Lim et al., 2022], and curriculum filtering. We differ in two ways: (i) we provide a spectral/energy-based explanation of *why* GNNs fit graph-level noise, rather than only mitigating it, and (ii) our practical method (GCOD) achieves comparable robustness at a fraction of the compute.

Dirichlet energy and over-smoothing. Dirichlet energy quantifies signal smoothness on graphs [Zhou and Schölkopf, 2005]. Existing analyses establish that GNNs act as low-pass filters whose energy decays exponentially with depth under spectral conditions [NT and Maehara, 2019,

89 Cai and Wang, 2020, Oono and Suzuki, 2020, Rusch et al., 2023], and that weight-matrix eigenvalues
90 control attraction/repulsion of node features [Giovanni et al., 2023]. Energy-based regularizers [Zhou
91 et al., 2021, Bo et al., 2021, Chen et al., 2023] address *too little* energy (over-smoothing) on *node*
92 classification. We address the opposite regime: *too much* within-graph energy caused by graph-level
93 label noise. Qian et al. [2025] recently show that, for graph classification specifically, over-smoothing
94 in *high-entropy* regions is the most damaging; we leverage this insight (Corollary 1) to argue why
95 suppressing energy growth is essential under graph-level noise. A more comprehensive related-work
96 discussion is in Appendix A.

97 3 Background

98 **Notation.** A graph is $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ with $N = |\mathcal{V}|$, degree matrix $\mathbf{D}$, adjacency $\mathbf{A} \in \{0, 1\}^{N \times N}$,
99 feature matrix $\mathbf{X} \in \mathbb{R}^{N \times m}$, combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq$
100 $\dots \leq \lambda_N$) and normalized Laplacian $\Delta = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$. We write $\lambda_2(\mathcal{G})$ for $\lambda_2(\mathbf{L})$ throughout
101 and use Δ only when explicitly stated. In *graph classification*, the dataset is $\mathcal{D} = \{(\mathcal{G}^i, \mathbf{y}_i)\}_{i=1}^n$,
102 $\mathbf{y}_i \in \{0, 1\}^{|C|}$ the one-hot graph label; we write c_i for the class. Under message passing, $\mathbf{H}_i^{l+1} =$
103 $\text{UP}_{\Omega_i}(\mathbf{H}_i^l, \text{AGGR}_{\mathbf{W}_i}(\mathbf{H}_i^l, \mathbf{A}^i))$, with $\mathbf{H}_i^0 \equiv \mathbf{X}^i$ and $\mathbf{Z}^i \equiv \mathbf{H}_i^L$. A permutation-invariant readout
104 $f_\theta : \mathbb{R}^{N_i \times m'} \rightarrow \mathbb{R}^{|C|}$ maps node features to class probabilities $\hat{\mathbf{y}}_i$. Under noisy labels, the observed
105 c_i may differ from the ground truth; the test set is clean.

106 **Dirichlet energy on graphs.** For graph $\mathcal{G}^i$ with representation $\mathbf{Z}^i \in \mathbb{R}^{N_i \times m'}$ we use the combinato-
107 rial Dirichlet energy

$$E^{\text{dir}}(\mathbf{Z}^i) = \sum_{(u,v) \in \mathcal{E}^i} \|\mathbf{Z}_u^i - \mathbf{Z}_v^i\|_2^2 = \text{tr}((\mathbf{Z}^i)^\top \mathbf{L}^i \mathbf{Z}^i). \quad (1)$$

108 Small E^{dir} means smooth (low-frequency) representations within the graph; large E^{dir} means sharp-
109 ening (high-frequency). The symmetrically-normalized variant $E_{\text{sym}}^{\text{dir}}(\mathbf{Z}) = \sum_{(u,v)} \|\mathbf{Z}_u / \sqrt{d_u} -$
110 $\mathbf{Z}_v / \sqrt{d_v}\|_2^2 = \text{tr}(\mathbf{Z}^\top \Delta \mathbf{Z})$, used in some over-smoothing analyses [NT and Maehara, 2019, Oono
111 and Suzuki, 2020], is equivalent up to degree-bounded constants and our qualitative claims hold for
112 either; we use (1) because it pairs cleanly with the Poincaré inequality of Section 6. To track behavior
113 across a class-restricted subset $\mathcal{S}$ of training graphs we use

$$E_{\text{set}}^{\text{dir}}(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{\mathcal{G}^i \in \mathcal{S}} E^{\text{dir}}(\mathbf{Z}^i). \quad (2)$$

114 Crucially, (1) is computed on *learned representations*, not on labels. Within-graph homophily of
115 features therefore evolves with training, decoupled from the graph-level label of $\mathcal{G}^i$.

116 4 When GNNs Fit Graph-Label Noise

117 GNNs trained with standard cross-entropy (CE) on graph classification tasks are not uniformly
118 vulnerable to label noise. On the full ogbg-ppa [Szklarczyk et al., 2018a] (37 classes, the standard
119 OGB benchmark), even 40% symmetric noise barely degrades CE accuracy, because the model
120 is under-parameterized for the full task and behaves robustly [Zhang et al., 2021]. To expose the
121 noise-fitting regime, we control three axes: **(F1)** number of classes (proxy for over-parameterization),
122 **(F2)** graph order (size of each graph), and **(F3)** training-set size. We use a 5-layer GIN [Xu et al.,
123 2019] with 300 hidden units throughout, unless stated.

124 In all three regimes (Fig. 1 and Appendix D), GNNs progressively memorize noisy graph-level
125 labels. This narrows the question to: what changes in the network’s internal representations during
126 memorization, and can this change be controlled?

127 5 Dirichlet Energy as a Diagnostic for Noise Fitting

128 We track $E_{\text{set}}^{\text{dir}}(\mathcal{D}_c)$, the within-graph energy averaged over training graphs of class c . Across architec-
129 tures and datasets (Fig. 2), the diagnostic is consistent: $E_{\text{set}}^{\text{dir}}$ falls during the early “learning” phase,

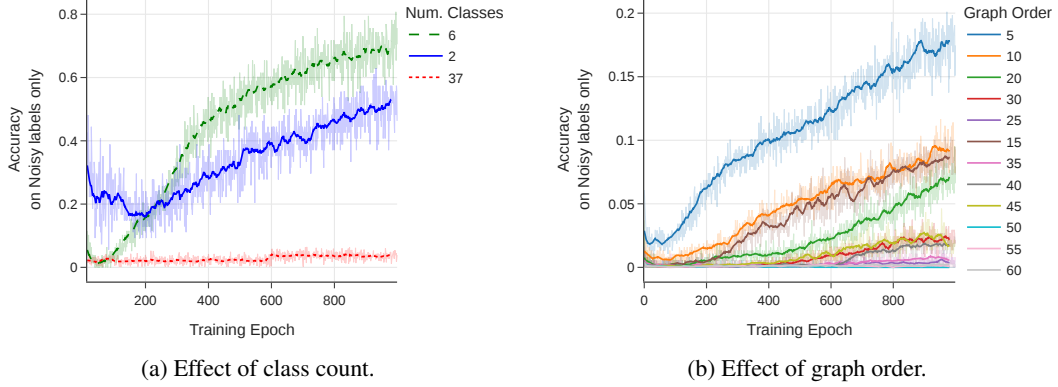


Figure 1: Training accuracy on *noisy labels only*: a higher curve means more memorization. (a) Reducing the number of PPA classes (with 20% symmetric noise) makes a fixed GIN over-parameterized, accelerating noise fitting. (b) Smaller average graph order (synthetic data, 35% noise) leaves less internal structure for averaging, enabling memorization.

then rises precisely when the model begins to fit noisy graphs (CE-20% in Fig. 2b), with 40% noise yielding a strictly larger end-of-training energy than 10% noise (Appendix F).

Energy is sensitive to noise specifically through high-frequency components. A natural concern is that energy also rises during clean-data overfitting [Yan et al., 2022]; we confirm this in Appendix F but show the noise-driven surge is markedly steeper. To localize the surge spectrally, we use the HLFF-GNN decomposition [Xu et al., 2024] to split representations into low-frequency (Z_1) and high-frequency (Z_2) components. On ENZYMES with 30% noise, $E^{\text{dir}}(Z_1)$ remains essentially unchanged, while $E^{\text{dir}}(Z_2)$ rises monotonically, and slope/variance statistics of $E^{\text{dir}}(Z_2)$ provide an early warning for label noise (Appendix E). HLFF-GNN is used purely as an analytical tool; the same total-energy surge appears in standard GIN/GCN.

Why energy is more useful than a generic overfitting signal. Standard noise-detection signals require a *clean* held-out set: validation loss is uninformative when validation labels follow the same noise distribution as training (the noisy validation loss continues to decrease as the model fits noise, as we observe in our setting); train-test gap is unobservable at training time; small-loss heuristics need a calibration regime. Within-graph Dirichlet energy is computed entirely on the training graphs and their representations, so it remains a valid noise-fitting indicator under fully-noisy supervision. Beyond replacing the missing clean signal, it tells you *how* the network is failing: which spectral components drive the surge. This mechanistic specificity is what turns a diagnostic into a design principle, and motivates the three interventions in Section 7, each of which targets the same harmful spectral behavior in a distinct way.

6 Why a Non-Trivial Energy Decrease Prevents Noise Fitting

We now show that, under standard regularity assumptions, the within-graph Dirichlet energy must grow when a GNN is forced to fit graph-level label noise, and that this growth is absorbed disproportionately by graphs with richer structure. The argument has three pieces: a dataset-level spectral identity (Proposition 1), the Graph Poincaré inequality (Lemma 1), and a bi-Lipschitz lower bound on the GNN (Lemma 2) [Chuang and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024]. Together they yield Theorem 1, which says that fitting noisy graph-level labels mathematically forces $E_{\text{set}}^{\text{dir}}$ to rise.

6.1 Three Building Blocks

Proposition 1 (Dataset-level Dirichlet energy). *Let $\mathcal{D} = \{\mathcal{G}^1, \dots, \mathcal{G}^n\}$ with representations $\mathbf{Z}^i \in \mathbb{R}^{N^i \times m'}$ and adjacencies $\mathbf{A}^i$. Define the block-diagonal graph $\bar{\mathcal{G}} = (\bar{\mathbf{Z}}, \bar{\mathbf{A}})$ with $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \parallel \dots \parallel \mathbf{Z}^n] \in \mathbb{R}^{N_{\text{tot}} \times m'}$ (vertical concatenation) and $\bar{\mathbf{A}} = \text{blkdiag}(\mathbf{A}^1, \dots, \mathbf{A}^n)$, with associated*

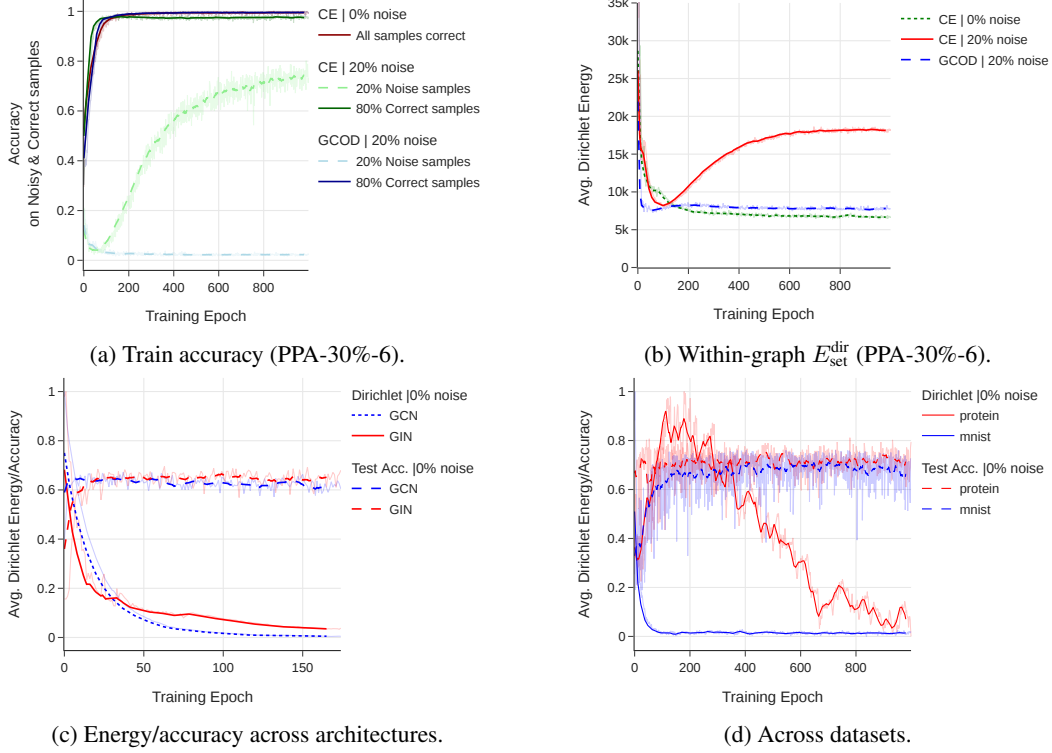


Figure 2: Within-graph Dirichlet energy of learned representations *rises during noise memorization*, across architectures (GIN/GCN) and datasets (PPA, MUTAG, MNIST, PROTEINS). Under clean labels, $E_{\text{set}}^{\text{dir}}$ steadily decays as accuracy improves; under noise, it follows the same decay until the model starts fitting noisy graphs, after which it sharply rises. GCOD suppresses this surge.

162 combinatorial Laplacian $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$. Then

$$E_{\text{set}}^{\text{dir}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}}) = \frac{1}{|\mathcal{D}|} \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2, \quad (3)$$

163 where $\{(\bar{\lambda}_u, \bar{\psi}_u)\}$ are eigenpairs of $\bar{\mathbf{L}}$, and the spectrum decomposes as $\bar{\mathbf{L}} = \bigcup_i \Lambda^i$ with each
 164 eigenvector supported on a single connected component $\mathcal{G}^i$.

165 The proof (Appendix B) follows from $\det(\bar{\mathbf{L}} - \lambda \mathbf{I}) = \prod_i \det(\mathbf{L}^i - \lambda \mathbf{I})$, since $\bar{\mathbf{L}}$ is block diagonal.
 166 The crucial structural fact is that *each high-frequency eigenvector of $\bar{\mathbf{L}}$ is owned by exactly one graph*;
 167 sharpening along such an eigenvector raises only that graph’s energy.

168 **Lemma 1** (Graph Poincaré, vector-valued form). *For any feature matrix $\mathbf{Z} \in \mathbb{R}^{N \times m'}$ on a connected*
 169 *graph $\mathcal{G}$ with combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ and second-smallest eigenvalue $\lambda_2(\mathcal{G}) > 0$, let*
 170 $\bar{\mathbf{z}} = \frac{1}{N} \sum_v \mathbf{Z}_v$. Then

$$\sum_{v=1}^N \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} E^{\text{dir}}(\mathbf{Z}). \quad (4)$$

171 (4) bounds within-graph variance of node features by the combinatorial Dirichlet energy scaled by the
 172 spectral gap. Proof: standard Courant–Fischer applied per feature dimension (Appendix B). The same
 173 statement holds with Δ in place of $\mathbf{L}$ provided $\mathbf{Z}$ is replaced by its degree-normalization $\mathbf{D}^{-1/2} \mathbf{Z}$
 174 throughout; we choose the form above so that the bound is on the variance of the raw representations.

175 **Lemma 2** (Bi-Lipschitz lower bound for a GIN block with 2-layer combine, adapted from Chuang and
 176 Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024). *Let $h_\theta(\mathbf{X}) = \sigma_2(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1) \mathbf{W}_2)$*
 177 *be a single GIN aggregation $\mathbf{M} = (1 + \varepsilon)\mathbf{I} + \mathbf{A} \in \mathbb{R}^{N \times N}$ followed by a two-layer MLP with*

178 weights $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$ and $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$ acting on the feature dimension, with the widening
 179 regime $m \leq m' \leq m''$ (so the linear maps have full column rank). Assume $\sigma_{\min}(\mathbf{M}) \geq \kappa > 0$ (e.g.,
 180 $1 + \varepsilon > |\lambda_{\min}(\mathbf{A})|$), $\sigma_{\min}(\mathbf{W}_k) \geq m_k > 0$, and that σ_1, σ_2 are leaky-ReLU with slope $\alpha > 0$ (or
 181 ReLU operated with a strict margin $\delta > 0$ along the line segment $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$). Then

$$\|h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F \geq c_{\text{lower}} \|\Delta\mathbf{X}\|_F, \quad c_{\text{lower}} = \alpha^2 m_1 m_2 \kappa. \quad (5)$$

182 A second aggregation step (turning the block into a true two-layer GIN $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$)
 183 replaces c_{lower} by $\alpha^2 m_1 m_2 \kappa^2$; the qualitative content of the lemma is unchanged.

184 The lemma is the singular-value chain rule on the vectorized Jacobian $\mathbf{J} = \mathbf{D}_2(\mathbf{W}_2^\top \otimes \mathbf{I}_N) \mathbf{D}_1(\mathbf{W}_1^\top \otimes$
 185 $\mathbf{M})$, where $\mathbf{D}_k = \text{diag}(\sigma'_k)$ and the Kronecker structure follows from $\text{vec}(\mathbf{A}\mathbf{Y}\mathbf{B}) = (\mathbf{B}^\top \otimes$
 186 $\mathbf{A})\text{vec}(\mathbf{Y})$; on the regularity regime above, $\sigma_{\min}(\mathbf{J}) \geq c_{\text{lower}} > 0$, and (5) follows by the fundamental
 187 theorem of calculus along γ . The role of the lemma is to express that the GIN does not collapse
 188 meaningful feature differences, the property the model uses (and abuses) when fitting noisy labels.

189 6.2 Main Theorem: Noise Fitting Forces Energy Growth

190 Consider two graphs $\mathcal{G}^i, \mathcal{G}^j$ with the *same* number of nodes $N = N^i = N^j$ and structurally similar
 191 pooled descriptors $\mathbf{g}_i \approx \mathbf{g}_j$ on the clean training distribution. (The same- N assumption is mainly for
 192 notational simplicity; the qualitative conclusion extends to the general case via vertex-permutation
 193 matching, see Remark below.) Suppose label noise flips c_j to $c'_i \neq c_i$, so the network must produce
 194 *different* graph-level outputs for them. Let $\mathbf{h}_i \in \mathbb{R}^{N \times m'}$ denote the per-node post-message-passing
 195 signal of $\mathcal{G}^i$, and $\bar{\mathbf{h}}_i = \mathbf{1}_N \mathbf{g}_i^\top$ its node-wise mean replicated to a constant signal at $\mathbf{g}_i = \frac{1}{N} \sum_v \mathbf{h}_{i,v}$.
 196 **Theorem 1** (Graph-level noise fitting forces E^{dir} to grow). *Under the hypotheses of Lemmas 1–2*
 197 *and $N^i = N^j = N$,*

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \leq \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}. \quad (6)$$

198 In particular, if the noisy training objective combined with an L -Lipschitz readout forces a graph-level
 199 output gap of at least Δ_{out} , so that $\|\mathbf{h}_i - \mathbf{h}_j\|_F \geq \Theta_{\text{noise}} := L^{-1} \Delta_{\text{out}}$, while the structural similarity
 200 gives $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$, then

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \geq \Theta_{\text{noise}} - \sqrt{N} \varepsilon. \quad (7)$$

201 **Why this is non-trivial.** (7) is a *lower bound* on the within-graph Dirichlet energies forced by
 202 the noisy training objective. For two structurally similar graphs that must be mapped to different
 203 labels, $\mathbf{g}_i \approx \mathbf{g}_j$ at the pooled stage; the gap Θ_{noise} that the network must produce to fit the noise
 204 cannot come from the means alone. By the Poincaré inequality, the only remaining route is the
 205 within-graph variance, which is precisely E^{dir}/λ_2 . Energy growth is therefore a *mathematical*
 206 *consequence* of graph-level label noise fitting, not an empirical coincidence. The role of the bi-
 207 Lipschitz condition (Lemma 2) is to ensure that this required output gap cannot be obtained “for free”
 208 by collapsing different inputs onto the same representation: $c_{\text{lower}} > 0$ guarantees that maintaining
 209 $\|\mathbf{h}_i - \mathbf{h}_j\|_F \geq \Theta_{\text{noise}}$ at convergence is consistent with non-degenerate weights, and so the inequality
 210 has content. The non-triviality of the link in graph classification is exactly that this bound has no
 211 analog in node classification, where labels live on the same nodes whose smoothness E^{dir} measures.

212 **Remark 1** (Unequal graph sizes). When $N^i \neq N^j$, replace $\|\mathbf{h}_i - \mathbf{h}_j\|_F$ in (6) by $\inf_\pi \|\mathbf{h}_i - \pi(\mathbf{h}_j)\|_F$
 213 where π ranges over zero-padded permutation matchings. The triangle inequality and per-graph
 214 Poincaré bounds carry through unchanged for each graph; only the cross-term $\sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2$ is
 215 replaced by $\sqrt{\min(N^i, N^j)} \|\mathbf{g}_i - \mathbf{g}_j\|_2$ plus a residual $|N^i - N^j| \cdot \|\mathbf{g}_\bullet\|_2^2$ from the padded rows.
 216 The energy lower bound (7) therefore continues to hold with $\sqrt{N} \rightarrow \sqrt{\min(N^i, N^j)}$ and a shifted
 217 Θ_{noise} .

218 6.3 Asymmetric Burden on High-Entropy Graphs

219 (7) only constrains the *sum* of $\sqrt{E^{\text{dir}}(\mathbf{Z}^i)/\lambda_2(\mathcal{G}^i)}$ across the two graphs. We next show that the
 220 structural-entropy ranking of Qian et al. [2025] dictates which graph absorbs the burden.

For a graph $\mathcal{G}$, the structural Shannon entropy is $H(\mathcal{G}) = -\sum_i \frac{d_i}{\sum_j d_j} \log \frac{d_i}{\sum_j d_j}$. Theorems 1–2 of Qian et al. [2025] prove that more nodes (at fixed sparsity) and denser connectivity (at fixed $|V|$) both strictly increase H .

Lemma 3 (Spectral counting). *The combinatorial Laplacian $\mathbf{L}^i$ has at most N^i distinct eigenvalues, contained in $[0, \max_{(u,v) \in \mathcal{E}^i} (d_u + d_v)]$, hence in $[0, 2d_{\max}(\mathcal{G}^i)]$ [Chung, 1997]. Equivalently, the normalized spectrum $\Lambda(\Delta^i) \subset [0, 2]$ with $\lambda_{\max}(\Delta^i) = 2$ iff $\mathcal{G}^i$ has a bipartite component. Therefore, when sharpening lifts spectral mass onto eigenvectors $\bar{\psi}_u$ of $\bar{\mathbf{L}}$ with $\bar{\lambda}_u \geq \tau$ for some threshold τ , the share allocated to $\mathcal{G}^i$ is bounded above by the count $n^i(\tau) = |\{j : \lambda_j^i \geq \tau\}|$, which is non-decreasing in N^i and in connection density (by Cauchy interlacing under vertex/edge addition).*

Corollary 1 (Asymmetric burden). *If $H(\mathcal{G}^j) \geq H(\mathcal{G}^i)$ via either of the conditions in Qian et al. [2025, Thm. 1–2], then the count of high-frequency eigenvalues of $\mathcal{G}^j$ above any threshold τ is at least that of $\mathcal{G}^i$, so the energy increase forced by Theorem 1 is preferentially absorbed by $\mathcal{G}^j$. This collapses the node-distribution distinctness of $\mathcal{G}^j$ [Qian et al., 2025, Sec. 3.3], the very property pooled graph classifiers rely on.*

Implication. Theorem 1 says noise fitting must raise energy somewhere; Corollary 1 says it raises it preferentially in graphs the classifier most needs to distinguish. *Suppressing within-graph energy growth therefore blocks the noise-fitting path while preserving the classification signal*, an alignment that has no analog in node classification. This is precisely the non-trivial property of graph classification we set out to establish.

Remark 2. This explains why, in Section 5, the surge of $E^{\text{dir}}(Z_2)$ under noise is much steeper than under clean overfitting. Clean overfitting can be absorbed across the full spectrum and across both means and variances; noise fitting on similar graphs is structurally constrained to high-frequency, high-entropy directions.

7 Three Convergent Methods

The theory of Section 6 identifies three distinct levers to suppress harmful within-graph energy growth: the *spectrum of the weight matrices* (architectural), the *energy of representations* (regularization), and the *loss function* (sample re-weighting). We design one intervention for each, and use them as convergent evidence for the energy framework.

7.1 Method 1: Spectral Constraint on Weight Matrices (CE+W2)

Giovanni et al. [2023] show that, in linear graph convolutions with *symmetric* weight matrices $\mathbf{W} = \mathbf{Q}\mathbf{A}\mathbf{Q}^\top$, the gradient flow of the Dirichlet energy decomposes along the eigenvectors of $\mathbf{W}$: components corresponding to positive λ_k evolve under attraction (smoothing) of connected nodes, while components with negative λ_k undergo repulsion (sharpening). Together with Proposition 1, this argues for projecting the post-aggregation matrix $\mathbf{W}_l^2$ onto its positive eigenspace whenever it is symmetric.

When $\mathbf{W}_l^2$ is non-symmetric, the GRAFF dichotomy no longer applies cleanly: eigenvalues may be complex and the flow has rotational components without a sign-based smoothing/sharpening interpretation. We therefore split $\mathbf{W}_l^2 = \mathbf{S}_l + \mathbf{N}_l$ with $\mathbf{S}_l = \frac{1}{2}(\mathbf{W}_l^2 + (\mathbf{W}_l^2)^\top)$ symmetric (real spectrum) and $\mathbf{N}_l = \frac{1}{2}(\mathbf{W}_l^2 - (\mathbf{W}_l^2)^\top)$ skew-symmetric (purely imaginary spectrum, hence no signed contribution to the GRAFF smoothing/sharpening dichotomy), and apply the projection only to $\mathbf{S}_l$. This is a heuristic extension of the GRAFF analysis, not a theorem; we validate it empirically via the energy curves of Fig. 2.

Procedure. After each gradient step, eigendecompose $\mathbf{S}_l = \Phi_l \mu_l \Phi_l^\top$ (μ_l real, Φ_l orthogonal), project $\mu_l \rightarrow [\mu_l]^+ = \max(\mu_l, 0)$, and reconstruct $\mathbf{W}_l^{2+} = \Phi_l [\mu_l]^+ \Phi_l^\top + \mathbf{N}_l$. The projection is detached from the gradient graph (Appendix C.4). **CE+W2** is CE training with this projection; it requires no clean validation data and no extra hyperparameters. It is most useful as a theoretical instantiation of the energy story; in practice, the per-epoch eigendecomposition is costly and can destabilize convergence (Appendix C.4), motivating the alternatives below.

269 7.2 Method 2: Explicit Dirichlet-Energy Regularization ($\mathcal{L}_{DE}$)

270 A more direct lever is to penalize $E^{\text{dir}}(\mathbf{Z}^i)$ above a threshold U_{c_i} that is class-aware:

$$\mathcal{L}_{DE}(\mathcal{D}) = \frac{1}{N} \sum_{i=1}^N [\max(0, E^{\text{dir}}(\mathbf{Z}^i) - U_{c_i})]^2, \quad \mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{DE}. \quad (8)$$

271 We instantiate the threshold in two ways. **Class-specific:** U_c is recomputed every epoch as the
 272 mean E^{dir} over a small clean validation set restricted to class c . This adapts to class-dependent
 273 baseline energies (e.g. a small molecular graph’s natural energy is different from a social graph’s),
 274 but requires a clean held-out set. **Fixed:** $U_c = U$ globally, a single hyperparameter that does not
 275 require any clean validation data, at the cost of mild over-smoothing on clean labels. Both variants
 276 improve robustness; the class-specific variant additionally improves *clean-label* accuracy by acting
 277 as a generalization-aware regularizer.

278 7.3 Method 3: Graph Centroid Outlier Discounting (GCOD)

279 GCOD is our practical recommendation. It extends NCOD [Wani et al., 2023] (originally an
 280 image-classification loss) to graph classification, with two design changes that are necessary in the
 281 graph-classification regime: a third KL term that disentangles clean from noisy graphs via per-sample
 282 trainable parameters $\mathbf{u} \in [0, 1]^n$, and feedback through the current training accuracy a_{train} that
 283 prevents premature outlier discounting. For batch B with predictions $f_\theta(\mathbf{Z}_B)$, hard predictions $\hat{\mathbf{y}}_B$,
 284 soft labels $\tilde{\mathbf{y}}_B$ (centroid-based, see Appendix C.6), and observed labels $\mathbf{y}_B$:

$$\mathcal{L}_1 = \mathcal{L}_{CE}(f_\theta(\mathbf{Z}_B) + a_{\text{train}} \text{diag}_{\text{vec}}(\mathbf{u}_B) \mathbf{y}_B, \tilde{\mathbf{y}}_B), \quad (9)$$

$$\mathcal{L}_2 = \frac{1}{|C|} \|\hat{\mathbf{y}}_B + \text{diag}_{\text{vec}}(\mathbf{u}_B) \mathbf{y}_B - \mathbf{y}_B\|^2, \quad (10)$$

$$\mathcal{L}_3 = (1 - a_{\text{train}}) D_{KL}\{\mathfrak{L}, \sigma(-\log \mathbf{u}_B)\}, \quad (11)$$

285 with $\mathfrak{L} = \log \sigma(\text{diag}_{\text{mat}}(f_\theta(\mathbf{Z}_B) \mathbf{y}_B^\top))$. Updates are split: $\theta \leftarrow \theta - \alpha \nabla_\theta (\mathcal{L}_1 + \mathcal{L}_3)$, $\mathbf{u} \leftarrow \mathbf{u} - \beta \nabla_{\mathbf{u}} \mathcal{L}_2$.
 286 Empirically, GCOD reduces $E_{\text{set}}^{\text{dir}}$ during training (Fig. 2) without explicitly minimizing it: by down-
 287 weighting graphs flagged as noisy via large u_i , it removes the only gradient pressure that would push
 288 the network toward high-frequency directions in the first place. GCOD *is not* a theoretically derived
 289 energy minimizer. Instead, the theory predicts that any sample-level mechanism that suppresses
 290 noise-driven gradients will reduce $E_{\text{set}}^{\text{dir}}$, which is exactly the convergent evidence we observe.

291 8 Experiments

292 **Setup.** We evaluate on ogbg-ppa [Szklarczyk et al., 2018a] (full and a controlled 30%/6-class
 293 subset), MUTAG, PROTEINS, ENZYMES, IMDB-BINARY, REDDIT-MULTI-12K, MSRC-21,
 294 MNIST-graph, and Cora (node classification, transfer test). Backbones are GIN [Xu et al., 2019] and
 295 GCN [Kipf and Welling, 2017]; full hyperparameters are in Appendix C. Baselines: standard CE;
 296 SOP [Liu et al., 2022], a strong noise-robust loss from image classification; OMG [Yin et al., 2023],
 297 the leading graph-classification noise method. All numbers are mean $\pm$ std over 4 runs.

298 **Why a controlled PPA subset?** On the full 37-class PPA, all methods are under-parameterized for
 299 the task, and CE itself is robust (Section 4); a noise-robustness method cannot help where the baseline
 300 does not fail. We use a 30%/6-class PPA subset as a *controlled* testbed where over-parameterization
 301 triggers noise fitting, and we additionally report the full-PPA result to show our methods do no harm
 302 there. Generality claims are grounded in the seven additional benchmarks.

303 8.1 Main Result: Controlled PPA Subset

304 Table 1 reports the full ablation on the controlled PPA subset. GCOD is best at every noise level
 305 and, crucially, has the smallest gap between best and final test accuracy, indicating that it does
 306 not overfit late in training. CE+W2 and $\mathcal{L}_{DE}$ both improve robustness over CE, confirming the
 307 convergent-evidence prediction of Section 6. Under clean labels, $\mathcal{L}_{DE}^*$ even outperforms CE, a direct
 308 effect of the class-aware energy threshold.

Table 1: Test accuracy on the PPA-30%-6 subset (5-layer GIN, mean $\pm$ std over 4 runs). **Best** is highlighted in red and bold, second-best in blue. “CE clean only” shows accuracy on the clean subset of the noisy data, as a reference upper bound. Class-specific $\mathcal{L}_{DE}^*$ requires a clean validation set.

Noise	Method	Test Acc.		Train Acc.	
		Best	Last	Best	Last
0%	CE	96.25 $\pm$ 0.05	91.25	100.0	99.33
	GCOD	96.65 $\pm$ 0.52	93.25	99.23	99.03
	CE+W2	96.50 $\pm$ 1.12	86.50	99.76	99.29
	$\mathcal{L}_{DE}$ (Fixed)	96.58 $\pm$ 0.27	85.70	99.26	98.29
	$\mathcal{L}_{DE}^*$ (Class-spec.)	96.96 $\pm$ 0.04	88.67	99.41	98.67
20%	CE (clean only)	96.15 $\pm$ 0.09	90.66	84.50	84.07
	CE	88.66 $\pm$ 0.16	62.58	94.98	93.45
	SOP	91.01 $\pm$ 0.44	85.50	79.17	77.64
	GCOD	93.91 $\pm$ 0.26	92.58	80.11	79.52
	CE+W2	89.83 $\pm$ 1.23	76.66	84.90	83.68
	$\mathcal{L}_{DE}$ (Fixed)	89.69 $\pm$ 0.57	80.25	78.07	70.14
	$\mathcal{L}_{DE}^*$	92.34 $\pm$ 0.41	80.92	84.55	83.73
40%	CE (clean only)	95.08 $\pm$ 0.13	80.44	68.15	67.05
	CE	82.08 $\pm$ 0.22	51.83	82.47	78.88
	SOP	82.33 $\pm$ 1.32	65.41	58.90	57.68
	GCOD	93.88 $\pm$ 0.04	91.08	65.09	64.31
	CE+W2	88.25 $\pm$ 1.13	56.33	68.78	65.88
	$\mathcal{L}_{DE}$ (Fixed)	88.58 $\pm$ 0.75	77.12	61.08	54.53
	$\mathcal{L}_{DE}^*$	88.55 $\pm$ 0.11	71.83	68.98	67.80

Table 2: GCOD matches or beats CE across diverse datasets. Mean over 4 runs; $\pm$ std reported where available. Sym_{20} : 20% symmetric noise. $Asym_{40}$: 40% asymmetric noise. The $Asym_{40}$ GCOD column reports the std deviations from [Wani et al., 2023]’s rebuttal protocol.

Dataset	0% CE	Sym_{20} CE	Sym_{20} GCOD	$Asym_{40}$ CE	$Asym_{40}$ GCOD
PROTEINS	81.16	76.23	79.38	72.90	76.19 $\pm$ 1.22
MNIST	72.69	66.30	71.26	71.15	72.61 $\pm$ 0.43
ENZYMES	73.33	64.16	68.69	65.80	69.81 $\pm$ 0.98
IMDB-B	76.50	75.00	75.40	68.18	72.89 $\pm$ 0.91
MUTAG	94.73	89.47	90.01	91.16	93.19 $\pm$ 0.38
REDDIT	48.15	45.05	44.98	48.01	47.94 $\pm$ 2.87
MSRC-21	96.69	90.26	94.69	94.39	95.57 $\pm$ 0.32

309 8.2 Generalization Across Datasets and Noise Types

310 Table 2 reports GCOD across seven benchmarks beyond PPA. GCOD consistently improves over
311 CE under both 20% symmetric and 40% asymmetric noise, and matches CE on REDDIT where the
312 baseline is already saturated by class imbalance. Section 8 of Theorem 1’s prediction (energy growth
313 scales with the demanded Θ_{noise}) is consistent with the empirical observation that gains grow with
314 the noise rate (Appendix F).

315 **Full ogbg-ppa.** Under the full 37-class benchmark, all methods including CE behave robustly to
316 noise (Section 4); we therefore use it only as a do-no-harm check. Detailed results are in Appendix H.

Table 3: OMG protocol (Yin et al., 2023). Percentage gain over CE.

Dataset	OMG	GCOD
MUTAG	0.061	0.062
IMDB-B	0.047	0.049
PROTEINS	0.039	0.041

Table 4: Relative training time (CE-GIN = 1.00).

Method	Runtime
CE-GIN	1.00
SOP	1.048
GCOD	1.029
CE+W2	1.33
OMG [Yin et al., 2023]	≈ 3.00

Comparison with OMG. Under the OMG [Yin et al., 2023] protocol (Table 3), GCOD matches OMG’s accuracy (within 0.1–0.2 pp) while requiring $\sim 70\times$ less training overhead (Table 4): OMG adds $\approx 200\%$ runtime relative to CE-GIN, while GCOD adds only $\approx 2.9\%$. The accuracy-per-FLOP advantage is substantial and matters for any deployment beyond toy benchmarks.

Cross-setting transfer to node classification (Cora). As a stress test of generality, we apply GCOD and CE+W2 unchanged to node classification on Cora under 20% symmetric noise, against 13 node-specific baselines (NRGNN, RTGNN, GNN-Cleaner, ERASE, CR-GNN, and others; full table in Appendix I). On both GAT and GCN backbones, our methods rank in the top half despite not being designed for node classification: CE+W2 reaches 0.8157 (rank #2) on GAT and 0.7247 (rank #5) on GCN; GCOD reaches 0.7910 (rank #4) on GAT and 0.7403 (rank #4) on GCN. The fact that an energy-based regularizer that does not access neighbor labels or homophily can outperform specialized node-classification methods is direct evidence that the perspective captures a general property of GNN training under label noise rather than a graph-classification-specific quirk.

8.3 Convergent Evidence for the Energy Story

The three methods are mechanistically distinct, CE+W2 is architectural, $\mathcal{L}_{DE}$ is regularization, GCOD is sample re-weighting, yet they all (i) reduce $E_{\text{set}}^{\text{dir}}(\mathcal{D}_c)$ during training (Fig. 2 and Appendix F) and (ii) improve robustness on noisy graph classification. This convergence is the strongest causal evidence we can offer in absence of a fully closed-form theoretical characterization: three independent levers acting on the same target produce the same outcome. Theorem 1 and Corollary 1 together explain why this is expected.

9 Conclusion

We have argued, theoretically and empirically, that GNN robustness to graph-level label noise is governed by the within-graph Dirichlet energy of learned representations. The connection is non-trivial in graph classification, requiring an entropy-based argument that does not exist in node classification, and it leads naturally to three convergent interventions (CE+W2, $\mathcal{L}_{DE}$, GCOD) that all improve robustness across eight benchmarks while preserving clean-label performance. GCOD, our practical recommendation, requires no clean validation data and adds only $\approx 3\%$ training overhead, while matching the accuracy of state-of-the-art OMG at $\sim 70\times$ lower compute.

Limitations. Our theoretical bridge to graph classification, while non-trivial, is not yet a closed-form characterization of the noise-energy relationship. Real-world noisy graph-classification benchmarks are scarce; we follow the standard synthetic-noise protocol, which limits ecological validity. Extending the framework to instance-dependent noise and to non-classification graph tasks (regression, link prediction) is a natural direction.

Broader impact. Robust graph learning matters in chemistry and biology, where annotation errors are systematic. Methods that achieve robustness without requiring clean validation sets (GCOD) reduce the cost of deploying GNNs in such settings.

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552 A Additional Related Work

553 A.1 Learning under Label Noise

554 Beyond robust losses, additional families include sample relabelling [Arazo et al., 2019, Reed et al.,
555 2015], two-network co-teaching and divide-and-mix approaches [Han et al., 2018, Li et al., 2020,
556 Kim et al., 2023], and regularization through MixUp [Zhang et al., 2017]. Reweighting methods adapt
557 sample importance via training dynamics or held-out probes [Liu and Tao, 2015, Pleiss et al., 2020].
558 Most of these were developed for image classification; their direct transfer to graph classification is
559 non-obvious because graphs lack the strong feature regularities of images.

560 A.2 Graph Learning under Noise

561 Most graph-related noise work targets node classification, exploiting either homophily (RS-GNN [Dai
562 et al., 2022], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a], pairwise interactions [Du
563 et al., 2023]) or self-supervision (pseudo-label denoising [Yuan et al., 2023b, Qian et al., 2023]).
564 Edge/feature noise has also been studied, e.g. structural noise [Fox and Rajamanickam, 2019, Dai
565 et al., 2022]. These methods typically assume connected nodes share labels, which is incompatible
566 with our graph-classification setting where labels are graph-level. The seminal work of NT et al.
567 [2019] treats graph classification under label noise via a surrogate loss; OMG [Yin et al., 2023]
568 combines contrastive learning, MixUp, and curriculum filtering. Both are practical improvements but
569 provide no spectral or energy-based explanation of *why* GNNs fit graph-level noise.

570 A.3 Dirichlet Energy and Spectral Properties of GNNs

571 NT and Maehara [2019] and Rusch et al. [2023] analyze GNNs as low-pass filters, with self-
572 loops further shrinking eigenvalues. Cai and Wang [2020], Oono and Suzuki [2020] prove that
573 GNN Dirichlet energy decays exponentially when the product of the largest singular value of the
574 weight matrix and the largest Laplacian eigenvalue is below 1. Giovanni et al. [2023] disentangle
575 smoothing/sharpening via positive/negative weight-eigenvalues. Zhou et al. [2021] (NeurIPS) uses
576 energy regularization to prevent over-smoothing on *node* classification. Our setting is the opposite
577 regime: too *much* energy under graph-level label noise.

578 Qian et al. [2025] introduce a structural-entropy viewpoint on over-smoothing for graph classification,
579 showing that high-entropy regions are most damaged by over-smoothing. We invert and combine
580 this with Proposition 1 to get Corollary 1: the graphs that suffer most from over-*sharpening* (the
581 noise-fitting regime) are also the high-entropy graphs that carry the most discriminative signal. The
582 two papers therefore identify symmetric ends of the same spectral failure.

583 A.4 Lipschitz/Stability Analyses

584 Chuang and Jegelka [2022] bound GIN outputs via the Tree Mover’s Distance, and Davidson and
585 Dym [2025] extend uniform-Lipschitz analyses to MPNNs through Hölder continuity of aggrega-
586 tion/combination/readout. Juvina et al. [2024] derive optimal Lipschitz constants $\vartheta = \phi(\lambda_K)$ for
587 graph convolutions under non-negative weights and ReLU. Gama et al. [2019] show output stability
588 under graph-shift-operator perturbations. These results characterize stability w.r.t. inputs; we instead
589 study stability against *label* noise, mediated by the within-graph energy of learned representations.

590 B Proofs and Extended Theory

591 B.1 Proof of Proposition 1

592 Let $\Lambda^i = \{\lambda_u^i\}_{u=1}^{N^i}$ be the spectrum of the per-graph combinatorial Laplacian $\mathbf{L}^i = \mathbf{D}^i - \mathbf{A}^i$. Define
593 the block-diagonal Laplacian

$$\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n), \quad \det(\bar{\mathbf{L}} - \lambda \mathbf{I}_{N_{\text{tot}}}) = \prod_{i=1}^n \det(\mathbf{L}^i - \lambda \mathbf{I}_{N^i}), \quad (12)$$

so $\bar{\Lambda} = \bigcup_i \Lambda^i$. Let ψ_j^i be the j -th eigenvector of $\mathbf{L}^i$ at eigenvalue λ_j^i . The vector

$$\bar{\psi}_u = (0, \dots, 0, \psi_j^i, 0, \dots, 0)^\top \in \mathbb{R}^{N_{\text{tot}}} \quad (13)$$

(zero on every block other than i) satisfies $\bar{\mathbf{L}}\bar{\psi}_u = \lambda_j^i \bar{\psi}_u$. With $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \parallel \dots \parallel \mathbf{Z}^n]$ (vertical concatenation), the combinatorial Dirichlet energy of the block-diagonal graph satisfies

$$E^{\text{dir}}(\bar{\mathbf{Z}}) = \text{tr}(\bar{\mathbf{Z}}^\top \bar{\mathbf{L}} \bar{\mathbf{Z}}) = \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2 = \sum_i \sum_{r,j} \lambda_j^i (\psi_j^{i\top} \mathbf{Z}_{:,r}^i)^2 = \sum_i E^{\text{dir}}(\mathbf{Z}^i), \quad (14)$$

where the second equality uses the spectral expansion in the orthonormal basis $\{\bar{\psi}_u\}$ and the third uses the block-supported structure of each $\bar{\psi}_u$. Hence $E_{\text{set}}^{\text{dir}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_i E^{\text{dir}}(\mathbf{Z}^i) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}})$. $\square$

B.2 Proof of Lemma 1 (Graph Poincaré Inequality)

The combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is real symmetric and positive semi-definite. Because $\mathcal{G}$ is connected, $\ker(\mathbf{L}) = \text{span}(\mathbf{1})$, and the eigenvalues are $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$. By the Courant–Fischer min–max characterization,

$$\lambda_2(\mathcal{G}) = \min_{\substack{\mathbf{x} \perp \mathbf{1} \\ \mathbf{x} \neq \mathbf{0}}} \frac{\mathbf{x}^\top \mathbf{L} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}. \quad (15)$$

Fix any feature dimension r and let $\mathbf{z} = \mathbf{Z}_{:,r} \in \mathbb{R}^N$ with $\bar{z} = \frac{1}{N} \sum_v z_v$. The centered vector $\mathbf{z} - \bar{z}\mathbf{1}$ is orthogonal to $\mathbf{1}$, so

$$\lambda_2(\mathcal{G}) \|\mathbf{z} - \bar{z}\mathbf{1}\|_2^2 \leq (\mathbf{z} - \bar{z}\mathbf{1})^\top \mathbf{L} (\mathbf{z} - \bar{z}\mathbf{1}) = \mathbf{z}^\top \mathbf{L} \mathbf{z} = \sum_{(u,v) \in \mathcal{E}} (z_u - z_v)^2, \quad (16)$$

using $\mathbf{L}\mathbf{1} = \mathbf{0}$. The right-hand side is the per-feature-dimension contribution to $E^{\text{dir}}(\mathbf{Z})$ in (1). Summing over $r = 1, \dots, m'$ yields

$$\sum_{v=1}^N \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 = \sum_{r=1}^{m'} \|\mathbf{z}^{(r)} - \bar{z}^{(r)}\mathbf{1}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} \sum_{r=1}^{m'} \mathbf{z}^{(r)\top} \mathbf{L} \mathbf{z}^{(r)} = \frac{E^{\text{dir}}(\mathbf{Z})}{\lambda_2(\mathcal{G})}, \quad (17)$$

which is (4). The constant $1/\lambda_2$ is sharp, attained by the Fiedler vector ψ_2 . The corresponding bound for the normalized Dirichlet energy $E_{\text{sym}}^{\text{dir}}(\mathbf{Z}) = \text{tr}(\mathbf{Z}^\top \Delta \mathbf{Z})$ is obtained by the change of variable $\mathbf{y}_v = \mathbf{Z}_v / \sqrt{d_v}$ and applying the same argument with $\lambda_2(\mathbf{L})$ in the bound. $\square$

B.3 Proof of Lemma 2 (Bi-Lipschitz Lower Bound for a GIN)

The block under analysis is one GIN aggregation followed by a two-layer MLP combine,

$$h_\theta(\mathbf{X}) = \sigma_2(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1) \mathbf{W}_2), \quad (18)$$

acting on $\mathbf{X} \in \mathbb{R}^{N \times m}$ with $\mathbf{M} \in \mathbb{R}^{N \times N}$ mixing rows, and $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$, $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$ acting on the feature (column) dimension. For leaky-ReLU activations (or ReLU with a fixed margin $\delta > 0$ along $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$, ensuring no piecewise-linear region change), each σ_k has derivative $\geq \alpha$ everywhere along the path.

Vectorized Jacobian. Using $\text{vec}(\mathbf{A}\mathbf{Y}\mathbf{B}) = (\mathbf{B}^\top \otimes \mathbf{A})\text{vec}(\mathbf{Y})$,

$$\begin{aligned} \text{vec}(\mathbf{M}\mathbf{X}\mathbf{W}_1) &= (\mathbf{W}_1^\top \otimes \mathbf{M}) \text{vec}(\mathbf{X}), \\ \text{vec}(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1) \mathbf{W}_2) &= (\mathbf{W}_2^\top \otimes \mathbf{I}_N) \mathbf{D}_1 (\mathbf{W}_1^\top \otimes \mathbf{M}) \text{vec}(\mathbf{X}), \end{aligned}$$

so the full Jacobian in vectorized form is

$$\mathbf{J}(\mathbf{X}) = \mathbf{D}_2 (\mathbf{W}_2^\top \otimes \mathbf{I}_N) \mathbf{D}_1 (\mathbf{W}_1^\top \otimes \mathbf{M}), \quad (19)$$

with $\mathbf{D}_k = \text{diag}(\sigma'_k) \succeq \alpha \mathbf{I}$.

619 **Singular-value chain.** For any product $\mathbf{ABCD}$ where each factor has $\sigma_{\min} > 0$, $\sigma_{\min}(\mathbf{ABCD}) \geq$
620 $\sigma_{\min}(\mathbf{A})\sigma_{\min}(\mathbf{B})\sigma_{\min}(\mathbf{C})\sigma_{\min}(\mathbf{D})$ (chain rule for singular values; equivalent to applying the lower
621 bound $\|\mathbf{Ax}\| \geq \sigma_{\min}(\mathbf{A})\|\mathbf{x}\|$ inductively). Using $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$,

$$\begin{aligned}\sigma_{\min}(\mathbf{J}) &\geq \sigma_{\min}(\mathbf{D}_2) \sigma_{\min}(\mathbf{W}_2^\top \otimes \mathbf{I}_N) \sigma_{\min}(\mathbf{D}_1) \sigma_{\min}(\mathbf{W}_1^\top \otimes \mathbf{M}) \\ &\geq \alpha \cdot m_2 \cdot \alpha \cdot m_1 \kappa = \alpha^2 m_1 m_2 \kappa = c_{\text{lower}}.\end{aligned}$$

622 **Fundamental theorem of calculus.** For $\gamma(t) = \mathbf{X} + t \Delta \mathbf{X}$, under the activation regularity above
623 $h_\theta \circ \gamma$ has the same piecewise-linear region throughout, so

$$\text{vec}(h_\theta(\mathbf{X} + \Delta \mathbf{X}) - h_\theta(\mathbf{X})) = \left(\int_0^1 \mathbf{J}(\gamma(t)) dt \right) \text{vec}(\Delta \mathbf{X}). \quad (20)$$

624 Taking the ℓ_2 norm of both sides (= Frobenius norm in matrix form) and using $\|\mathbf{Jx}\|_2 \geq$
625 $\sigma_{\min}(\mathbf{J}) \|\mathbf{x}\|_2$ pointwise (so $\|\int \mathbf{J} dt \mathbf{x}\| \geq c_{\text{lower}} \|\mathbf{x}\|$ by Jensen on the singular-value lower bound),

$$\|h_\theta(\mathbf{X} + \Delta \mathbf{X}) - h_\theta(\mathbf{X})\|_F \geq c_{\text{lower}} \|\Delta \mathbf{X}\|_F. \quad (21)$$

626 For a true two-layer GIN $\sigma_2(\mathbf{M}\sigma_1(\mathbf{MXW}_1)\mathbf{W}_2)$, the second Kronecker factor in the chain becomes
627 $(\mathbf{W}_2^\top \otimes \mathbf{M})$ instead of $(\mathbf{W}_2^\top \otimes \mathbf{I}_N)$, contributing an extra $\sigma_{\min}(\mathbf{M}) \geq \kappa$ and giving $c_{\text{lower}} =$
628 $\alpha^2 m_1 m_2 \kappa^2$. $\square$

629 B.4 Proof of Theorem 1 (Noise Forces Energy Growth)

630 Let $\mathbf{h}_i, \mathbf{h}_j \in \mathbb{R}^{N \times m'}$ be the post-message-passing per-node signals of $\mathcal{G}^i$ and $\mathcal{G}^j$ (we assume $N^i =$
631 $N^j = N$; the unequal- N case is treated in the Remark in the main text). Let $\mathbf{g}_i = \frac{1}{N} \sum_v \mathbf{h}_{i,v} \in \mathbb{R}^{m'}$
632 be the mean-pooled graph descriptor, and let $\bar{\mathbf{h}}_i = \mathbf{1}_N \mathbf{g}_i^\top$ be its replication to a constant per-node
633 signal. By the Frobenius-norm triangle inequality,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \leq \|\mathbf{h}_i - \bar{\mathbf{h}}_i\|_F + \|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F + \|\bar{\mathbf{h}}_j - \mathbf{h}_j\|_F. \quad (22)$$

634 **Middle term.** Because both $\bar{\mathbf{h}}_i, \bar{\mathbf{h}}_j$ are constant signals of size $N \times m'$,

$$\|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F^2 = \sum_{v=1}^N \|\mathbf{g}_i - \mathbf{g}_j\|_2^2 = N \|\mathbf{g}_i - \mathbf{g}_j\|_2^2, \quad (23)$$

635 so $\|\bar{\mathbf{h}}_i - \bar{\mathbf{h}}_j\|_F = \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2$.

636 **Variance terms.** Identifying $\mathbf{h}_i$ with $\mathbf{Z}^i$ (the layer- L representation whose Dirichlet energy we
637 measure), Lemma 1 applied to $\mathbf{Z}^i$ gives

$$\|\mathbf{h}_i - \bar{\mathbf{h}}_i\|_F^2 = \sum_v \|\mathbf{h}_{i,v} - \mathbf{g}_i\|_2^2 \leq \frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}, \quad (24)$$

638 and analogously for $\mathcal{G}^j$. Taking square roots and combining,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \leq \sqrt{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}, \quad (25)$$

639 which is (6).

640 **Lower bound from the noisy training objective.** The graph-level prediction is $\hat{\mathbf{y}} = R(\mathbf{h})$ for some
641 permutation-invariant readout R that is L -Lipschitz in the Frobenius norm (mean- or sum-pooling
642 followed by an MLP with bounded weights). At convergence, the noisy training objective enforces
643 $\hat{\mathbf{y}}_i$ to lie near a one-hot vector at c_i and $\hat{\mathbf{y}}_j$ near a one-hot at $c_j \neq c_i$, so $\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq \Delta_{\text{out}}$ for some
644 $\Delta_{\text{out}} > 0$. By the readout's upper-Lipschitz property,

$$\|\mathbf{h}_i - \mathbf{h}_j\|_F \geq L^{-1} \|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq L^{-1} \Delta_{\text{out}} =: \Theta_{\text{noise}}. \quad (26)$$

645 The role of Lemma 2 is to make this lower bound non-vacuous: $c_{\text{lower}} > 0$ ensures that the encoder has
646 a positive lower-Lipschitz constant at convergence, so the noisy-label constraint cannot be satisfied
647 by collapsing $(\mathbf{X}^i, \mathbf{A}^i)$ and $(\mathbf{X}^j, \mathbf{A}^j)$ to the same internal representation; equivalently, the encoder
648 remains in a regularity regime where $\|\mathbf{h}_i - \mathbf{h}_j\|_F$ is a meaningful quantity.

649 **Combining upper and lower bounds.** Under the structural-similarity assumption $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$,
650 substituting $\Theta_{\text{noise}} \leq \|\mathbf{h}_i - \mathbf{h}_j\|_F$ into (6) and rearranging,

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \geq \Theta_{\text{noise}} - \sqrt{N} \varepsilon, \quad (27)$$

651 which is (7). $\square$

B.5 Proof of Lemma 3 (Spectral Counting)

The combinatorial Laplacian $\mathbf{L}^i$ is symmetric positive semi-definite. By Anderson–Morley’s bound, every eigenvalue λ_j^i of $\mathbf{L}^i$ satisfies $\lambda_j^i \leq \max_{(u,v) \in \mathcal{E}^i} (d_u + d_v) \leq 2d_{\max}(\mathcal{G}^i)$. Equivalently, the normalized Laplacian $\Delta^i = \mathbf{D}^{-1/2} \mathbf{L}^i \mathbf{D}^{-1/2}$ has eigenvalues in $[0, 2]$, with $\lambda_{\max}(\Delta^i) = 2$ iff $\mathcal{G}^i$ has a bipartite component [Chung, 1997, Lemma 1.7, Thm. 1.13]. The number of distinct eigenvalues of $\mathbf{L}^i$ is at most N^i .

For any threshold $\tau > 0$, define $n^i(\tau) = |\{j : \lambda_j^i \geq \tau\}|$. We claim $n^i(\tau)$ is non-decreasing under (a) edge addition and (b) vertex addition with at least one incident edge.

(a) *Edge addition*: if $\mathcal{G}^{i'} = \mathcal{G}^i + \{(u, v)\}$, then $\mathbf{L}^{i'} = \mathbf{L}^i + (\mathbf{e}_u - \mathbf{e}_v)(\mathbf{e}_u - \mathbf{e}_v)^\top$, a rank-one PSD update. By Weyl’s monotonicity theorem (Horn and Johnson, 2012, Cor. 4.3.12), $\lambda_k(\mathbf{L}^{i'}) \geq \lambda_k(\mathbf{L}^i)$ for every k . Hence if $\lambda_j^i \geq \tau$ then $\lambda_j^{i'} \geq \tau$, so $n^{i'}(\tau) \geq n^i(\tau)$.

(b) *Vertex addition*: write $\mathcal{G}^{i'}$ as the result of starting from the padded graph $\mathcal{G}^i \sqcup \{v_{N+1}\}$ (an isolated extra vertex) and then adding all incident edges $\{(v_{N+1}, u) : u \in \mathcal{N}(v_{N+1})\}$ one at a time. The padded Laplacian is $\mathbf{L}_0 = \mathbf{L}^i \oplus [0] \in \mathbb{R}^{(N^i+1) \times (N^i+1)}$, with spectrum $\Lambda(\mathbf{L}^i) \cup \{0\}$. Each subsequent edge addition is a rank-one PSD update covered by case (a), so $\lambda_k(\mathbf{L}^{i'}) \geq \lambda_k(\mathbf{L}_0)$ for all k . Since $\Lambda(\mathbf{L}_0) \setminus \{0\} = \Lambda(\mathbf{L}^i) \setminus \{0\}$, the count above any $\tau > 0$ satisfies $n^{i'}(\tau) \geq n_0(\tau) = n^i(\tau)$. (The case $\tau = 0$ is trivial since $n^{i'}(0) = N^i + 1 \geq N^i = n^i(0)$.)

Now let $\bar{\mathbf{L}} = \text{blkdiag}(\mathbf{L}^1, \dots, \mathbf{L}^n)$ as in Proposition 1. By the block-diagonal structure, every eigenpair $(\bar{\lambda}_u, \bar{\psi}_u)$ of $\bar{\mathbf{L}}$ is supported on exactly one $\mathcal{G}^i$. Sharpening that lifts spectral mass onto eigenvectors with $\bar{\lambda}_u \geq \tau$ therefore allocates a share to $\mathcal{G}^i$ that is bounded above by $n^i(\tau) / \sum_k n^k(\tau)$, which grows with N^i and connection density of $\mathcal{G}^i$. $\square$

B.6 Proof of Corollary 1 (Asymmetric Burden)

Suppose $H(\mathcal{G}^j) \geq H(\mathcal{G}^i)$ either via more nodes at fixed sparsity (Theorem 1 of Qian et al., 2025) or via denser connectivity at fixed $|V|$ (Theorem 2 of Qian et al., 2025). Lemma 3 then implies $n^j(\tau) \geq n^i(\tau)$ for every τ , where $n^i(\tau)$ counts eigenvalues of $\mathcal{G}^i$ above τ . Now consider any sharpening update of $\bar{\mathbf{Z}}$ that increases the spectral coefficients on eigenpairs with $\bar{\lambda}_u \geq \tau$. By Proposition 1, each such eigenpair lies entirely on one of $\mathcal{G}^i$ or $\mathcal{G}^j$, and the count of available high-frequency eigenvectors on $\mathcal{G}^j$ is at least that on $\mathcal{G}^i$. Hence on average a $\bar{\lambda}_u \geq \tau$ component is more likely to be supported on $\mathcal{G}^j$, and the resulting energy increase is preferentially absorbed by $\mathcal{G}^j$. By Section 3.3 of Qian et al. [2025], increasing $E^{\text{dir}}(\mathbf{Z}^j)$ along its high-frequency eigenvectors collapses the pairwise distinctness of node-feature distributions inside $\mathcal{G}^j$, the precise quantity that pooled graph classifiers rely on for separation. Hence high-entropy graphs lose discriminability faster than low-entropy graphs. $\square$

B.7 Why a Decrease of $E_{\text{set}}^{\text{dir}}$ Is Non-Trivial in Graph Classification

Three consecutive observations together justify the term “non-trivial”:

(i) Noise is injected at the graph-label level, not at the node-feature level. Therefore, the increase of $E^{\text{dir}}(\mathbf{Z}^i)$ during noise fitting cannot be a re-encoding of the noise itself; it must be a behavioral response of the network through training dynamics.

(ii) For a randomly-initialized network, $E_{\text{set}}^{\text{dir}}$ is independent of class labels by construction. Any correlation between $E_{\text{set}}^{\text{dir}}$ and labels emerges from the optimization process, not from the data.

(iii) By Theorem 1 and Corollary 1, the only optimization path that fits incorrect graph-level labels is to lift spectral mass into high-entropy directions of $\bar{\mathbf{L}}$, which are precisely the directions that a clean, generalization-driven network should preserve. Constraining $E_{\text{set}}^{\text{dir}}$ therefore selectively blocks the noise-fitting path while allowing the clean-learning path, an alignment that does *not* hold for node classification, where label noise corrupts the same nodes whose smoothness the energy measures, and where decreasing energy can erase the discriminative signal.

These three points collectively make the connection between energy reduction and noise robustness in graph classification a non-trivial property. The analogous statement in node classification reduces to “the model fits homophily, and homophily is what we measured.”

C Experimental Setting

C.1 Datasets

ogbg-ppa [Szkarczyk et al., 2018a]: 158,100 protein-association graphs across 37 taxonomic groups, derived from STRING [Szkarczyk et al., 2018b]; the standard OGB graph-property-prediction benchmark. Modified PPA-30%-6 takes 30% of training graphs from 6 selected classes, used as a controlled testbed. **PROTEINS** [Borgwardt et al., 2005]: 1,113 protein graphs; 2 classes; mean 39 nodes/graph. **ENZYMES** [Borgwardt et al., 2005]: 600 enzyme graphs across 6 EC top-level classes. **MSRC-21** [Neumann et al., 2016]: 563 image-segmentation graphs, 20 classes. **MUTAG** [Kriege and Mutzel, 2012]: 188 molecular graphs, 2 classes. **IMDB-BINARY** [Yanardag and Vishwanathan, 2015b]: 1,000 social graphs, 2 classes. **REDDIT-MULTI-12K** [Yanardag and Vishwanathan, 2015b]: 11,929 thread-graphs, 11 classes. **MNIST-graph**: 55,000 superpixel graphs over the MNIST images, 10 classes.

C.2 Synthetic Datasets for Failure-Mode F2

We sample average graph orders $\bar{N} \in \{5, 10, \dots, 60\}$, drawing actual node counts from $\text{Poisson}(\bar{N})$. Each dataset has 6 classes with 1,400 graphs/class, fixed average degree 2, edges sampled uniformly. Node features are $\mathcal{N}(\mu_c, 1.5)$ with $\mu_c = f(\text{class})$. Symmetric noise rate is fixed at 35%.

C.3 Architectures and Hyperparameters

Unless otherwise stated, we use a 5-layer GIN [Xu et al., 2019] or GCN [Kipf and Welling, 2017] with 300 hidden units, Adam ($\eta = 10^{-3}$, weight decay 10^{-4}), batch size 32, 200–1000 epochs, $\eta_u = 1$ for GCOD’s outlier parameter. Compute: NVIDIA RTX A6000.

C.4 Spectral Bias Implementation Details (CE+W2)

For each layer l , after a gradient update, decompose $\mathbf{W}_l^2 = \Phi_l^2 \mu_l^2 (\Phi_l^2)^{-1}$ and replace by $\Phi_l^2 [\mu_l^2]^+ (\Phi_l^2)^{-1}$, where $[\cdot]^+$ is element-wise ReLU on the eigenvalues. The projection is detached from the gradient graph. We apply it to the post-aggregation matrix $\mathbf{W}^2$ of every layer; applying it also to $\mathbf{W}^1$ marginally improves smoothing but degrades stability. Convergence variance under this scheme is shown in Fig. 8.

C.5 Class-Specific and Fixed Bounds for $\mathcal{L}_{DE}$

Class-specific: after every epoch, set $U_c = |\mathcal{D}_c^{\text{val}}|^{-1} \sum_{i \in \mathcal{D}_c^{\text{val}}} E^{\text{dir}}(\mathbf{Z}^i)$. The validation set must be clean to keep U_c class-specific. **Fixed:** $U_c = U$ globally. We progressively decrease U from a large initial value until accuracy under clean labels begins to drop; the final U is the largest value that achieves a noticeable energy ceiling without over-smoothing.

C.6 Design of GCOD

We adopt the centroid-based soft labels of Wani et al. [2023]: normalize $\phi(\mathbf{z}_i) = \mathbf{z}_i / \|\mathbf{z}_i\|$; per-class centroid $\phi_c = \bar{\mathbf{z}}_c / \|\bar{\mathbf{z}}_c\|$; soft target $\tilde{\mathbf{y}}_i = [\max(\phi(\mathbf{z}_i) \phi_c^\top, 0) \mathbf{y}_i]$. Equations (9)–(11) are then optimized as in the main text. The training-accuracy feedback in $\mathcal{L}_1$ and $\mathcal{L}_3$ is critical: without it, $\mathbf{u}$ dominates early in training and discounts samples before genuine patterns are learned (Fig. 7).

D Failure-Mode Experiments (Extended)

Additional plots for the F1–F3 failure modes are provided here for completeness, including effect-of-dataset-size for both PPA and synthetic datasets.

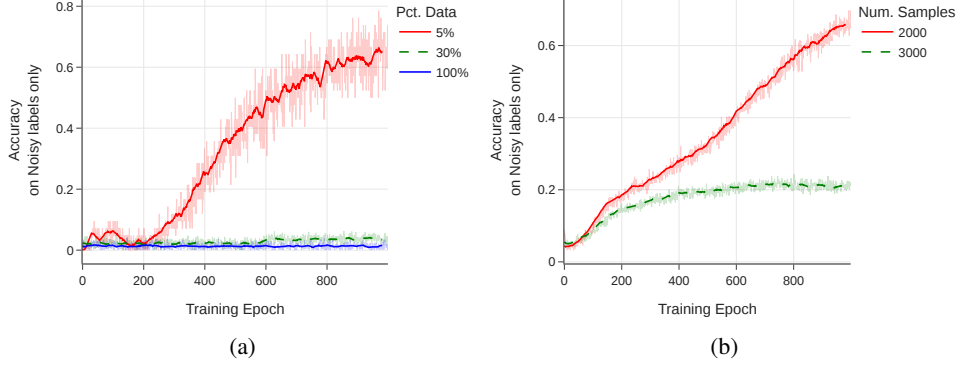


Figure 3: Effect of training-set size on noise memorization (training accuracy on noisy labels only). (a) PPA subsampling at 20% symmetric noise. (b) Synthetic graphs of order 7, 6 classes, 35% noise.

740 E Dirichlet-Energy Amplification in High-Frequency Components

741 We use the HLFF-GNN framework [Xu et al., 2024] (implemented as FGRLConv) to decompose
 742 representations into low-frequency Z_1 , high-frequency Z_2 , and residual Y . Under the composite loss

$$\mathcal{L} = \|Y - X\|_F^2 + \lambda \text{tr}(Z_1^\top \Delta Z_1) + \alpha \text{tr}(Z_2^\top (\mathbf{I} - \Delta) Z_2) + \beta (\|Y^\top Z_1\|_F^2 + \|Y^\top Z_2\|_F^2), \quad (28)$$

743 the spectral form of the high-frequency energy is $E^{\text{dir}}(Z_2) = \sum_{r,u} \lambda_u (\psi_u^\top Z_{2,r})^2$, in which large λ_u
 744 amplify any noise-driven content. On ENZYMES with 30% symmetric noise, we observe a sharp
 745 monotone increase in $E^{\text{dir}}(Z_2)$, while $E^{\text{dir}}(Z_1)$ and $E^{\text{dir}}(Y)$ remain stable, consistent with Theorem 1
 746 and Corollary 1.

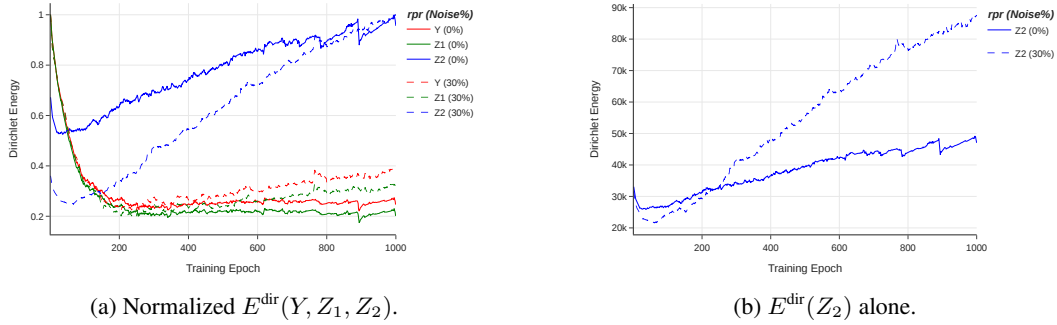


Figure 4: HLFF-GNN frequency decomposition on ENZYMES. Solid lines: clean labels. Dashed: 30% symmetric noise. The high-frequency component Z_2 is the only one whose energy diverges under noise, supporting the theoretical prediction.

747 F Additional Empirical Studies on Energy Behavior

748 **Energy under clean overfitting vs. noise.** On ENZYMES without regularization, a deliberately
 749 overfit GIN exhibits rising training accuracy and rising $E_{\text{set}}^{\text{dir}}$, confirming that energy tracks overfitting
 750 in general. However, the rise under noise is markedly steeper.

751 **Energy scales with noise rate.** On the controlled PPA subset, we sweep symmetric noise $\in$
 752 $\{10, 20, 30, 40, 60\}\%$. All curves follow the same U-shape (initial decrease, then rise), and the
 753 asymptotic energy increases monotonically with noise rate. This monotonicity is what makes $E_{\text{set}}^{\text{dir}}$ a
 754 diagnostic.

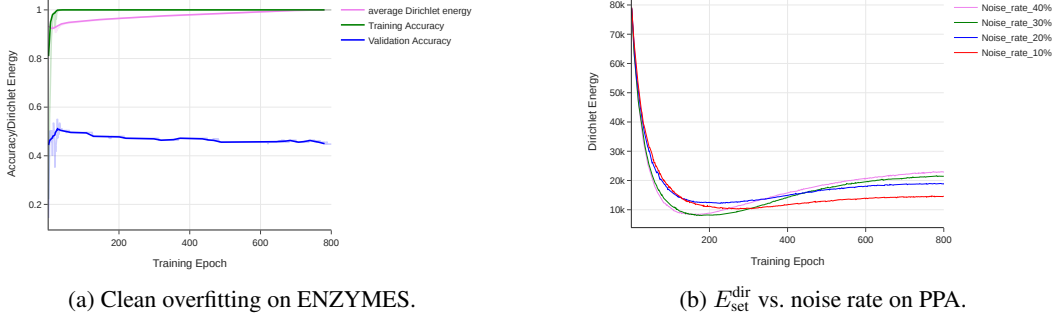


Figure 5: (a) Even without label noise, energy rises during overfitting; the surge under noise is much steeper. (b) Higher noise rates yield monotonically higher final energy.

Table 5: Sensitivity of GCOD to the learning rate η_u of the outlier parameter $\mathbf{u}$, under 40% asymmetric noise.

Dataset	$\eta_u = 1$	$\eta_u = 0.1$	% Change
PROTEINS	76.19	75.89	-0.39%
MNIST	72.61	72.48	-0.18%
ENZYMES	69.81	68.33	-2.12%
IMDB-B	72.89	71.94	-1.30%
MUTAG	93.19	92.61	-0.62%
REDDIT	47.94	48.08	+0.29%
MSRC-21	95.57	95.45	-0.13%

G Additional Tables and Figures

H Full ogbg-ppa: Do-No-Harm Check

The standard ogbg-ppa benchmark has 37 classes and $\sim 158,000$ graphs of mean order ≈ 240 , so a 5-layer GIN with 300 hidden units is structurally *under-parameterized* for the task. We trained CE, $\mathcal{L}_{DE}$ (Fixed), CE+W2 and GCOD on the full benchmark with 0%, 20% and 40% symmetric noise. Two qualitative observations are robust across runs:

1. *Training accuracy on the noisy-labeled subset stays close to chance for all methods, including standard CE.* The under-parameterized model never reaches a regime where it can memorize the corrupted graph-level labels, consistent with our failure-mode analysis of Section 4: F1 (number of classes), F2 (graph order) and F3 (training-set size) all push the full PPA out of the noise-fitting regime.
2. *Test accuracy of our three methods is statistically indistinguishable from CE* at every noise rate. None of CE+W2, $\mathcal{L}_{DE}$, or GCOD perturbs the underlying training dynamics enough to harm clean generalization, since their corrective signals (negative-eigenvalue projection, energy upper bound, outlier discounting) are only triggered by the high-frequency surge that does not occur on the full PPA.

This is by design: the methods are conditional regularizers, not unconditional smoothers. The take-away is that they impose *no measurable overhead in accuracy* when the underlying network is already noise-robust, while providing substantial protection when over-parameterization, low graph order, or small training sets push the network into the noise-fitting regime (Section 8, controlled PPA-30%-6 subset). A fair generality claim therefore rests on the seven additional benchmarks in Tables 2 and 6, with the full PPA serving exclusively as a do-no-harm sanity check.

Table 6: Performance of CE vs. GCOD with 20% symmetric label noise. Last column reports the absolute gain (GCOD–CE) on test accuracy.

Dataset	Metric	0% CE	20% CE	20% GCOD	20% (GCOD–CE)
MNIST	Best	72.69	66.30	71.26	+4.96
	Last	69.80	53.64	67.88	+14.24
	Diff	2.89	12.66	3.38	
ENZYMES	Best	73.33	64.16	68.69	+4.53
	Last	65.78	57.50	62.54	+5.04
	Diff	7.55	6.66	6.15	
MSRC-21	Best	96.69	90.26	94.69	+4.43
	Last	93.80	79.64	90.26	+10.62
	Diff	2.89	10.62	4.43	
PROTEINS	Best	81.16	76.23	79.38	+3.15
	Last	79.18	62.32	78.12	+15.80
	Diff	1.98	13.91	1.26	
MUTAG	Best	94.73	89.47	90.01	+0.54
	Last	84.21	68.42	86.84	+18.42
	Diff	10.52	21.05	3.17	
IMDB-B	Best	76.50	75.00	75.40	+0.40
	Last	71.60	71.00	73.50	+2.50
	Diff	4.90	4.00	1.90	
REDDIT	Best	48.15	45.05	44.98	−0.07
	Last	46.01	44.67	44.89	+0.22
	Diff	2.14	0.38	0.09	

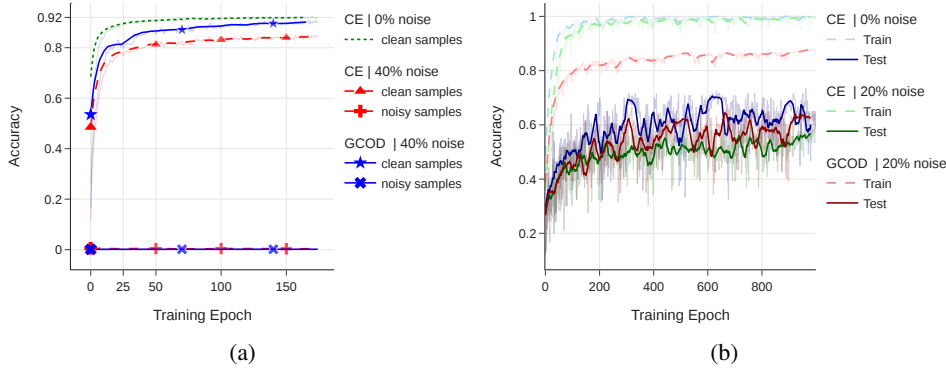


Figure 6: (a) Train accuracy on known noisy and clean PPA subsets (4-class, 40% symmetric noise) for GCN with CE: the model never separates noisy from clean training samples without a robust loss. (b) Train/test accuracy on ENZYMES with GIN under clean and 20% symmetric noise.

777 I Cross-Setting Transfer: Cora Node Classification

778 To assess whether the energy perspective captures a property of GNN training under label noise that
779 is more general than the graph-classification setting it was designed for, we apply CE+W2 and GCOD
780 *unchanged* to node classification on Cora at 20% symmetric label noise. We use both GAT and GCN
781 backbones, with the standard public split, and compare against 13 node-classification baselines that
782 include CE, GCE [Ghosh et al., 2017], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a],
783 GNN-Cleaner, ERASE, CR-GNN, and the standard noise-robust losses adapted to node classification.

784 Both methods land in the top half of the ranking on each backbone, despite using neither homophily,
785 contrastive losses, pseudo-labels, nor any node-specific machinery. CE+W2 in particular is competi-
786 tive with the strongest specialized node-classification baselines on GAT. The interpretation, consistent
787 with Theorem 1, is that the spectral mechanism by which noise fitting forces within-graph energy to
788 grow is not specific to graph classification: in node classification, individual labels are also embedded
789 in node-level representations whose Dirichlet energy reflects local smoothness, and constraining that
790 smoothness penalizes the same noise-fitting path. The transfer is therefore not a coincidence but a
791 corollary of the energy-based framing.

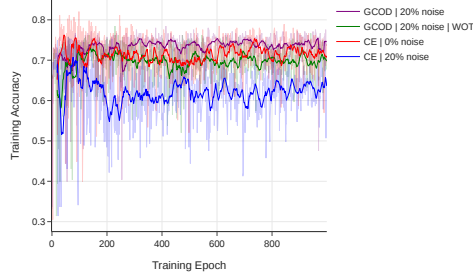


Figure 7: Ablation on the training-accuracy feedback in GCOD. Without scaling u by a_{train} (“GCOD WOT”), the outlier parameter dominates early and harms generalization; the full GCOD activates outlier-discounting gradually.

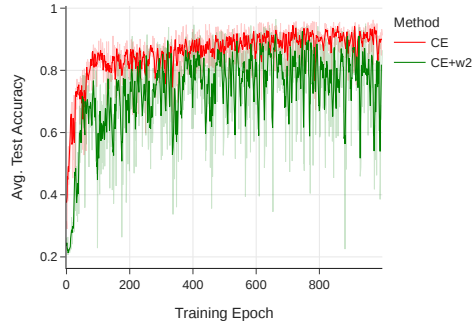


Figure 8: Test accuracy on PPA-30%-6 for CE and CE+W2. CE+W2 reaches comparable peak accuracy but exhibits unstable convergence due to the per-epoch eigendecomposition.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: The abstract and Section 1 state precisely the claims supported by the theoretical results in Section 6 and the experiments in Section 8.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: A dedicated paragraph in the Conclusion discusses the absence of a closed-form noise-energy characterization, reliance on synthetic noise, and the lack of standardized real-world noisy graph-classification benchmarks.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: Proposition 1, Lemmas 1–2, Theorem 1, Lemma 3 and Corollary 1 state their assumptions explicitly in the main text, and full proofs are provided in Appendix B.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

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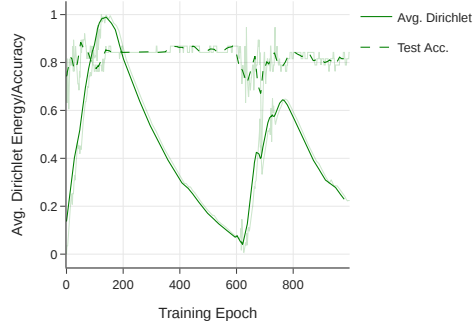


Figure 9: Average test accuracy and average within-graph $E_{\text{set}}^{\text{dir}}$ on MUTAG (0% noise) with GIN: classification accuracy improves while energy decays, the clean-data signature of a healthy GNN.

Table 7: Cross-setting transfer of GCOD and CE+W2 to node classification on Cora under 20% symmetric label noise. Numbers are test accuracy; ranks are with respect to the full 13-method comparison reported in the rebuttal materials. We report only the verified results for our two methods on each backbone; the “rank” field indicates their position in the full ranking. The methods are not modified, regularization parameters are kept at their graph-classification defaults.

Backbone	Method	Test Acc.	Rank (out of 13)
GAT	CE+W2	0.8157	#2
GAT	GCOD	0.7910	#4
GCN	GCOD	0.7403	#4
GCN	CE+W2	0.7247	#5

815 Justification: Appendix C provides datasets, splits, hyperparameters, architectures, and
816 the implementation details of all three methods. Anonymized code is provided in the
817 supplemental material.

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819 Question: Does the paper provide open access to the data and code, with sufficient instruc-
820 tions to faithfully reproduce the main experimental results?

821 Answer: [Yes]

822 Justification: All datasets are public benchmarks; an anonymous code repository accompa-
823 nies the submission.

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827 Answer: [Yes]

828 Justification: Section 8 and Appendix C report all training and evaluation details.

829 7. Experiment statistical significance

830 Question: Does the paper report error bars suitably and correctly defined or other appropriate
831 information about the statistical significance of the experiments?

832 Answer: [Yes]

833 Justification: All main results report mean $\pm$ std over 4 independent runs with different seeds.

834 8. Experiments compute resources

835 Question: For each experiment, does the paper provide sufficient information on the com-
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837 Answer: [Yes]

838 Justification: Appendix C reports the use of NVIDIA RTX A6000 GPUs and Table 4 reports
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867 Justification: The released code provides documentation for the proposed methods.

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881 Justification: No human subjects.

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